WEC 2026 data description

Main dataset for the contest

players_quarters_final.csv

(3 486 rows) – player activity snapshots at match quarter-hour checkpoints

This is the main analytical dataset for the contest.

It was constructed by aggregating event-level data from selected source tables (see below) – `player_appearance_shot_limited.csv` and `player_appearance_run.csv` into a single player-level panel organised around fixed time checkpoints within each match.

The dataset covers 31 matches and 869 unique player appearances. Each row represents one player at one time checkpoint, capturing their activity both in the preceding 15-minute window and cumulatively from the start of the match up to that point. The binary target variable **`scored_after`** indicates whether the player scored a goal at any point after the checkpoint in the same match, excluding own goals.

Observation unit and time structure. Checkpoints are defined within match periods, with minutes counted locally from the start of each period. Up to seven checkpoints are possible per match: minute 15, 30, and 45 of the first half (`H1_15`, `H1_30`, `H1_45`), minute 15 and 30 of the second half (`H2_15`, `H2_30`), and – for matches that reached extra time – minute 45 of the second half (`H2_45`) and minute 15 of the first extra time period (`ET1_15`). Only players who were actively on the pitch at the moment of the checkpoint are included, identified through the continuous `minute_in` and `minute_out` fields.

Feature groups. Variables with the prefix `last15_` describe the player's activity in the 15-minute window immediately preceding the checkpoint; variables with the prefix `cumul_` describe their accumulated activity from the start of the match. Both groups cover the same two domains: physical exertion (derived from `player_appearance_run.csv`) and shooting (derived from `player_appearance_shot_limited.csv`). Goals scored prior to the checkpoint were excluded from the shot features to prevent information leakage into the target variable.

Target variable. `scored_after` is a binary indicator equal to 1 if the player scored at least one non-own goal after the checkpoint in that match, and 0 otherwise. The target is heavily imbalanced: 203 positive cases out of 3,486 observations (approximately 5.8%), reflecting the natural rarity of goalscoring events in football.

Table 1. Description of columns in the main dataset
players_quarters_final.csv

Column	Description
`player_appearance_id`	Unique player-appearance identifier
`player_id`	Player identifier
`fixture_id`	Match identifier
`date`	Match date (`YYYY-MM-DD`)
`checkpoint`	Time checkpoint label: `H1_15`, `H1_30`, `H1_45`, `H2_15`, `H2_30`, `H2_45`, `ET1_15`
`checkpoint_period`	Match period of the checkpoint: `half_1`, `half_2`, `extra_time_1`
`checkpoint_min`	Local minute within the period at which the checkpoint is defined (15, 30, or 45)
`position`	Player position: `G` (goalkeeper), `D` (defender), `M` (midfielder), `A` (attacker)
`is_home`	Whether the player's team played at home (`TRUE/FALSE`)
`formation`	Tactical formation of the player's team (e.g. `4-1-4-1`, `3-4-3`)
`minute_in`	Minute the player entered the pitch, counted continuously from 1 across the full match
`minute_out`	Minute the player left the pitch, counted continuously from 1 across the full match
`subbed`	Whether the player was substituted on or off (`TRUE/FALSE`)
`jersey_number`	Shirt number
`last15_sprints`	Number of sprints (maximal-speed runs) in the last 15 minutes before the checkpoint
`last15_hsr`	Number of high-speed runs (sub-maximal) in the last 15 minutes before the checkpoint
`last15_distance`	Total distance covered in high-intensity runs (m) in the last 15 minutes
`last15_mean_max_speed`	Mean of per-run maximum speeds (m/s) in the last 15 minutes
`last15_peak_speed`	Highest maximum speed recorded across all runs (m/s) in the last 15 minutes
`last15_shots`	Number of shots (excluding own goals) in the last 15 minutes

Column	Description
`last15_shots_on_target`	Number of shots on target (goal or saved) in the last 15 minutes
`last15_shots_under_press`	Number of shots taken while under opponent pressure in the last 15 minutes
`last15_shots_top_third`	Number of shots originating from the attacking third in the last 15 minutes
`cumul_sprints`	Cumulative number of sprints from match start to the checkpoint
`cumul_hsr`	Cumulative number of high-speed runs from match start to the checkpoint
`cumul_distance`	Cumulative distance covered in high-intensity runs (m) from match start to the checkpoint
`cumul_mean_max_speed`	Mean of per-run maximum speeds (m/s) from match start to the checkpoint
`cumul_peak_speed`	Highest maximum speed recorded across all runs (m/s) from match start to the checkpoint
`cumul_shots`	Cumulative number of shots (excluding own goals) from match start to the checkpoint
`cumul_shots_on_target`	Cumulative number of shots on target from match start to the checkpoint
`cumul_shots_under_press`	Cumulative number of shots taken under pressure from match start to the checkpoint
`cumul_shots_top_third`	Cumulative number of shots from the attacking third from match start to the checkpoint
`scored_after`	**Target variable**: 1 if the player scored at least one goal after the checkpoint in this match, 0 otherwise (own goals excluded)

Source and supplementary datasets

for potential extension of feature space

player_appearance_pass.csv
(29 795 rows) – individual pass events

This table is the most granular passing log in the dataset, with one row per pass attempt made by a player during a match. Across nearly 30,000 rows, it captures every time a player chose to play the ball to a teammate — or tried to.

Each record is anchored to a specific player appearance via `player_appearance_id`, meaning passes can always be traced back to who made them, which team they played for, and in which match. When a pass has an intended recipient, the `addressee_player_appearance_id` field links to that receiver's appearance record — but this can be null, which likely indicates a clearance, a speculative long ball, or a situation where no specific target was identifiable.

The `accurate` flag is the core quality signal: it marks whether the pass actually reached its intended target. Combined with the total number of rows per player, this allows straightforward computation of pass volume and pass accuracy at any desired level of aggregation (per player, per time interval, per pitch zone).

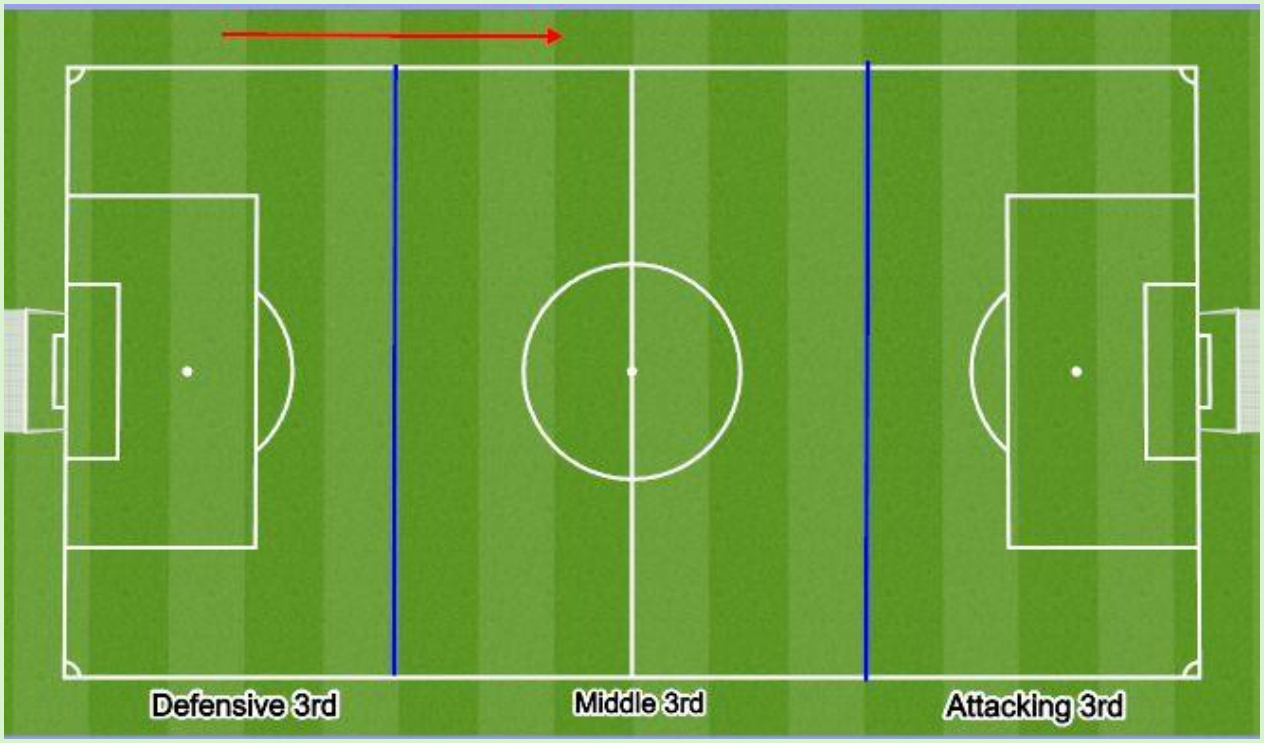
The table provides two key contextual dimensions. `period` tells us *when* in the match the pass occurred — distinguishing the first and second halves as well as both extra time periods — while `stage` encodes *where on the pitch* the pass originated, using a three-zone model: `bottom` (defensive third), `middle` (central area), and `top` (attacking third). The `minute` field adds finer temporal resolution within each period. Together, these allow analysis of how a player's passing behaviour evolves across the match and shifts across different areas of the pitch.

This table was NOT used to create features included in the main dataset for the contest.

Table 2. Description of columns in the supplementary dataset **player_appearance_pass.csv**

Column	Description
`id`	Unique pass event identifier (integer PK)
`period`	Match period: `half_1`, `half_2`, `extra_time_1`, `extra_time_2`
`player_appearance_id`	FK to `player_appearance.id` – the passer
`addressee_player_appearance_id`	FK to `player_appearance.id` – the intended receiver (NULL if no target)
`accurate`	Whether the pass reached its target (`True`/`False`)
`minute`	Minute of the match when the pass occurred
`stage`	Pitch zone where the pass originated: `bottom`, `middle`, `top`, or empty string

Figure 1. Illustration of pitch zones (stages)



`bottom` (defensive third) `middle` (central area) `top` (attacking third)

player_appearance_behaviour_under_pressure.csv
(12 185 rows) – passes made while being pressed

Where the pass table captures all passing events, this table zooms in on a specific and tactically rich subset: moments when a player receives and attempts to play the ball while being actively pressed by an opponent. With over 12,000 rows, it reflects how frequently pressing occurs in these matches and how players respond when under duress.

The structure shares several fields with the pass table — `period`, `minute`, `stage`, `accurate`, and the passer/receiver appearance IDs — but adds three fields that make this table uniquely informative. First, `pressing_player_appearance_id` identifies *who* applied the pressure, enabling analysis from the pressing player's perspective as well as the pressed player's. Second, `press_induced_outcome` captures what actually happened as a result of the press: the player may have played a `forward_pass`, `backward_pass`, or `lateral_pass`, carried the ball (`ball_carry`), or lost possession entirely (`turnover`). This makes it possible to measure not just whether a player was accurate under pressure, but whether the press succeeded in disrupting the direction of play. Third, `pass_angle` records the angle of the resulting pass in degrees — a continuous variable that is null when the outcome was a turnover or ball carry (i.e. no pass was made). This opens the door to geometric analysis of how pressure distorts passing direction.

Together, these fields make this one of the richest tables in the dataset for tactical analysis. It can answer questions like: which players consistently play forward under pressure? Which positions are pressed most often? Does pressing in the attacking third yield more turnovers than pressing in the defensive third?

This table was NOT used to create features included in the main dataset for the contest.

Table 3. Description of columns in the supplementary dataset **player_appearance_behaviour_under_pressure.csv**

Column	Description
`id`	Unique event identifier (integer PK)
`period`	Match period: `half_1`, `half_2`, `extra_time_1`, `extra_time_2`
`player_appearance_id`	FK to `player_appearance.id` – the player under pressure
`addressee_player_appearance_id`	FK to `player_appearance.id` – intended receiver (NULL if turnover/no target)
`accurate`	Whether the pass was accurate (`True`/`False`)
`pressing_player_appearance_id`	FK to `player_appearance.id` – the player applying pressure
`press_induced_outcome`	Outcome of the press: `forward_pass`, `backward_pass`, `lateral_pass`, `ball_carry`, `turnover`
`pass_angle`	Angle of the resulting pass in degrees (NULL when not applicable, e.g. turnover)
`minute`	Minute of the match
`stage`	Pitch zone: `bottom`, `middle`, `top`, or empty string

player_appearance_run.csv
(35 133 rows) – high-speed run / sprint events

This table captures the physical exertion dimension of each player's match, logging every high-intensity running effort they make. With over 35,000 rows — the largest event table in the dataset — it reflects just how frequently players make explosive movements during a match.

Each row represents a single running bout, classified by `run_type` into two categories: `hsr` (high-speed running, intense but sub-maximal effort) and `sprint` (maximal or near-maximal speed). This distinction is standard in sports science and reflects meaningfully different physical demands on the body. The table records both the `min_speed` and `max_speed` reached during each run in metres per second, as well as the total `distance` covered — allowing computation of peak output, average intensity, and cumulative physical load per player per match.

Contextual fields mirror those in the other event tables: `period` and `minute` place each run in time, while `stage` places it on the pitch. There is also a `possession` identifier, which links runs to specific possession sequences — a detail that enables more advanced analysis, such as whether players make more high-speed runs during their team's own attacks versus defensive transitions. This makes the table valuable not just for physical load analysis but also for understanding the tactical contexts in which players make their most intense efforts.

This table was used to create some features included in the main dataset for the contest, but they can be potentially further extended.

Table 4. Description of columns in the supplementary dataset **player_appearance_run.csv**

Column	Description
`id`	Unique run event identifier (integer PK)
`period`	Match period: `half_1`, `half_2`, `extra_time_1`, `extra_time_2`
`stage`	Pitch zone: `bottom`, `middle`, `top`
`possession`	Possession sequence identifier (integer)
`run_type`	Type of run: `hsr` (high-speed run) or `sprint`
`minute`	Minute of the match
`min_speed`	Minimum speed during the run (m/s, decimal)
`max_speed`	Maximum speed during the run (m/s, decimal)
`distance`	Distance covered during the run (meters, decimal)
`player_appearance_id`	FK to `player_appearance.id`

player_appearance_shot_limited.csv
(780 rows) – individual shot events

This table logs every shot attempt across all matches in the dataset, with one row per shot event. At 780 rows it is considerably smaller than the passing and running tables – reflecting the natural scarcity of shooting moments in football – but it is arguably the most information-dense table in the dataset, packing a rich set of contextual descriptors around each attempt.

Each shot is linked to the shooter via `player_appearance_id`, placing it within a specific player's appearance in a specific match. The `period` and `minute` fields situate the attempt in time, while `stage` places it on the pitch using the same three-zone model (`bottom`, `middle`, `top`) used throughout the dataset. The `possession` identifier connects each shot to a broader attacking sequence, which makes it possible to study how moves develop before they culminate in an attempt.

Where this table really stands out is in the detail it captures about *how* each shot was taken and *why* it ended the way it did. `body_part` records whether the attempt was made with the right foot, left foot, head, or another body part – a basic but important dimension of shot quality. `technique` goes further, distinguishing normal strikes from volleys, lobs, and overhead kicks, each of which carries very different expectations of accuracy and power. `play_pattern` describes how the attacking move was constructed: whether it came from regular build-up play, a counter-attack, a set piece (corner, direct or indirect free kick), a penalty, or even a throw-in – giving a full picture of the tactical context in which the shot arose.

The `outcome` field is the central variable in the table, recording what actually happened: the shot may have resulted in a `goal`, been `on_target` (saved), gone `off_target`, been `blocked`, or struck the `hit_woodwork`. Two additional outcomes – `own_goal` and `disallowed_goal` – cover edge cases, and the table handles these carefully through dedicated foreign keys. `own_goal_player_appearance_id` identifies the defending player who inadvertently scored, and is null for all non-own-goal events. Similarly, `block_player_appearance_id` identifies the player who blocked the shot, and is null whenever the shot was not blocked. This design means both the shooter's and the blocker's or own-goal scorer's perspectives can be recovered from a single row. **Note that `outcome` is excluded from the version of this table provided for the contest. Because the target variable `scored_after` in the main dataset is derived directly from whether a shot resulted in a goal, including `outcome` as a feature would constitute information leakage – a model trained on it would effectively be given the answer rather than learning to predict it.**

Finally, the `under_pressure` flag ties this table back to the pressing theme that runs through the dataset: it marks whether the shooter was being actively pressed at the moment of the attempt, enabling direct comparison of shot quality and outcome rates between pressured and unpressured situations.

This table was used to create some features included in the main dataset for the contest, but they can be potentially further extended.

Table 5. Description of columns in the supplementary dataset **player_appearance_shot_limited.csv**

Column	Description
`id`	Unique shot event identifier (integer PK)
`period`	Match period: `half_1`, `half_2`, `extra_time_1`, `extra_time_2`
`player_appearance_id`	FK to `player_appearance.id` – the shooter
`body_part`	Body part used: `right_foot`, `left_foot`, `head`, `other`
`technique`	Shot technique: `normal`, `volley`, `lob`, `overhead_kick`, `other`
`play_pattern`	How the attack was built: `regular_play`, `counter_attack`, `corner_kick`, `direct_free_kick`, `indirect_free_kick`, `penalty`, `throw_in`
`own_goal_player_appearance_id`	FK to `player_appearance.id` – the player who scored an own goal (NULL otherwise)
`block_player_appearance_id`	FK to `player_appearance.id` – the player who blocked the shot (NULL if not blocked)
`minute`	Minute of the match
`possession`	Possession sequence identifier (integer)
`stage`	Pitch zone: `bottom`, `middle`, `top`
`under_pressure`	Whether the shooter was under pressure (`True`/`False`)